

Bayesian multimodeling

About the course

- The focus is on the models, especially on the complex compositions of the models
 - ▶ Model criteria
 - ▶ Model complexity
 - ▶ Model search space
 - ▶ What is model optimality? Suboptimality?
 - ▶ How to optimize hyperparameters and metaparameters? What's the difference?
 - ▶ How to combine models?
 - ▶ How to operate with different heterogeneous models?

Course overlapping

- Bayesian model selection: deeper course with focus on math and details
- Generative models: focus on generative models

Topics in this term

- Distributions, expectation, likelihood
- Bayesian inference
- MDL
- Probabilistic metric spaces
- Variational inference ¹
- Informative prior, Sampling

¹+ bonus lecture on combination of optimization operators and ELBO by Andrey Veprikov

Scores

$$\text{Score} = \min(10, \text{round}(2 + (2 * \text{Forms} + 7 * \text{Project} + 1 * \text{Publication bonus})) \cdot \text{Paper review})$$

- **Paper review**

- ▶ Every class there will be 1-2 talks from students
- ▶ This year: not only a talk, but also a blog post describing the paper in details.
- ▶ No scores for students who didn't give at least one talk.

- Forms are **very** simple and done just to have a backoff for the minimal student grade.²

²Credits: Anastasia Voznyuk

Projects

Student projects are devoted to reproduce multiple well-established papers and wrap them into an open-source library. For the projects you can make a team up to 4 people (but it means you must implement at least 4 different papers in the project). Approximate schedule of the project can be found at the course page.

Expected results from the team:

- An implemented library
- A **short** blog-post message
- A tech. report

For the brave ones we can try to publish the results in the peer-reviewed journal (+1 point to all the team).

The list of projects and details will be presented next week.

An example of the good project result is Relaxit: <https://github.com/intsystems/relaxit>.

Talk

- Timing: 10-15 minutes
- Structure:
 - ▶ Title
 - ▶ Task/model motivation
 - ▶ Formal problem statement
 - ▶ Theory and method description
 - ▶ Experiments, examples, applications
 - ▶ Literature
 - ▶ Two simple questions (will be inserted into the form)
- For poorly done talks the score is zero.

This year you are also encouraged to write a blog-post paper explaining the paper in details. The style of the text is presuming the qualified audience (students of the 2-3rd year of our department).

Each blog post must be cross-reviewed by another student, the revision must be done after the review.

Homework

- Decide certainly if you attend the course
- Fill the form (the link will be on the page course)
- Have a look at libraries we might work with during the project:
 - ▶ JAX (and all the JAX ecosystem)
 - ▶ Pyro
 - ▶ (for enthusiasts) Julia
- Check that you have rights for intsystems orgainziation at github
- For those in charge: check you are able to record classes onto youtube and rutube

Page: <https://github.com/intsystems/BMM>
TG: see page :)